

M1M3 Thermal Control Optimization: Gradient Penalty Analysis and Rate Limit Selection

Rubin Observatory Thermal Control Team

February 2026

Abstract

This report presents a simulation study of the M1M3 mirror thermal control system, focusing on the tradeoff between temperature tracking performance and internal thermal gradients. We develop a two-zone thermal model that captures surface-bulk temperature dynamics and evaluate three operational scenarios across a range of setpoint rate limits. Results indicate that the current rate limit of $0.7^{\circ}\text{C}/\text{h}$ produces peak internal gradients of 0.6°C to 0.8°C , depending on the assumed gradient relaxation time constant. We provide recommendations for rate limit selection based on acceptable gradient thresholds and identify critical parameters requiring empirical validation.

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1 Introduction

The Rubin Observatory M1M3 primary/tertiary mirror requires active thermal control to maintain optimal image quality during nighttime observations. The mirror must be kept slightly cooler than ambient air (target: 0.3°C below ambient) to prevent “mirror seeing”—convective plumes arising from a warm mirror surface that degrade the point spread function.

However, rapid temperature changes introduce a competing concern: **internal thermal gradients**. The M1M3 mirror’s honeycomb structure creates complex thermal response patterns. When the HVAC system drives rapid setpoint changes, the mirror surface responds faster than the bulk material, creating temperature gradients that can cause:

- Differential thermal expansion and mechanical stress
- Surface figure distortions
- Focal plane image granularities from non-uniform thermal response

This report quantifies the tradeoff between tracking performance (minimizing mirror-ambient temperature difference) and gradient safety (minimizing internal temperature gradients) through simulation.

2 Thermal Model Description

2.1 Single-Zone Baseline Model

The baseline thermal control system models the mirror as a single thermal mass with first-order dynamics:

$$\frac{dT_{\text{mirror}}}{dt} = \frac{T_{\text{setpoint}} - T_{\text{mirror}}}{\tau} \quad (1)$$

where $\tau = 3.0\text{ h}$ is the mirror thermal time constant. The setpoint is rate-limited:

$$\left| \frac{dT_{\text{setpoint}}}{dt} \right| \leq r_{\text{max}} \quad (2)$$

where r_{max} is the maximum allowable rate of change (currently 0.7°C/h).

2.2 Two-Zone Gradient Model

To capture internal gradient dynamics, we developed a two-zone model that tracks both surface and bulk temperatures:

$$\frac{dT_{\text{surface}}}{dt} = \frac{T_{\text{setpoint}} - T_{\text{surface}}}{\tau_{\text{surface}}} \quad (3)$$

$$\frac{dT_{\text{bulk}}}{dt} = \frac{T_{\text{surface}} - T_{\text{bulk}}}{\tau_{\text{bulk}}} \quad (4)$$

$$\text{gradient} = T_{\text{surface}} - T_{\text{bulk}} \quad (5)$$

2.2.1 Model Parameters

Table 1: Two-zone thermal model parameters

Parameter	Symbol	Value	Description
Surface time constant	τ_{surface}	1.0 h	HVAC-to-surface response
Bulk time constant	τ_{bulk}	3.0 h	Surface-to-bulk conduction
Gradient relaxation	τ_{gradient}	2.0 h–3.0 h	Internal equilibration time
Simulation timestep	Δt	0.25 h	15-minute intervals

The gradient relaxation time constant τ_{gradient} is a critical uncertain parameter. Based on the mirror’s physical characteristics:

- Height gradient across mirror: 0.5 °C to 1.0 °C typical
- Honeycomb structure introduces thermal pattern complexity
- Estimated range: $\tau_{\text{gradient}} = 2.0 \text{ h to } 3.0 \text{ h}$

3 Simulation Methodology

3.1 Dataset

Simulations were performed using 392 nights from the Cerro Pachon temperature database (December 2025 – January 2026), using odd-date nights as the test set.

3.2 Parameter Sweep

We evaluated performance across:

- Maximum rate: $r_{\text{max}} \in \{0.3, 0.5, 0.7, 1.0, 1.25, 1.5, 2.0\} \text{ °C/h}$
- Gradient time constant: $\tau_{\text{gradient}} \in \{2.0, 2.5, 3.0\} \text{ h}$

3.3 Performance Metrics

1. **Tracking RMS:** Root-mean-square error between mirror temperature and target (ambient -0.3 °C) during overnight observations
2. **Percentage within tolerance:** Fraction of overnight period with $|T_{\text{mirror}} - T_{\text{target}}| < 0.3 \text{ °C}$
3. **Peak gradient:** Maximum $|T_{\text{surface}} - T_{\text{bulk}}|$ observed
4. **Mean gradient:** Average gradient magnitude over observation period
5. **Gradient integral:** Accumulated $\int |\text{gradient}|^2 dt$ (damage metric)

4 Results

4.1 Tracking Performance vs. Rate Limit

Table 2 summarizes tracking performance as a function of the maximum setpoint rate.

Table 2: Tracking performance metrics vs. maximum rate limit

Max Rate ($^{\circ}\text{C}/\text{h}$)	% Within $\pm 0.3^{\circ}\text{C}$	RMS Error ($^{\circ}\text{C}$)	Sunset Error ($^{\circ}\text{C}$)	Sunset Std ($^{\circ}\text{C}$)
0.30	35.0	0.81	-0.03	0.07
0.50	48.5	0.57	-0.05	0.09
0.70	58.6	0.44	-0.08	0.11
1.00	69.1	0.34	-0.12	0.12
1.25	74.4	0.30	-0.15	0.12
1.50	78.2	0.27	-0.17	0.13
2.00	82.6	0.23	-0.20	0.14

Note: Highlighted row indicates current operating point.

Key finding: Tracking performance improves monotonically with increased rate limit. The current $0.7^{\circ}\text{C}/\text{h}$ achieves 58.6% within tolerance; increasing to $1.0^{\circ}\text{C}/\text{h}$ would improve this to 69.1%.

4.2 Internal Gradient Results

Table 3: Peak and mean internal gradients vs. rate limit and τ_{gradient}

Max Rate ($^{\circ}\text{C}/\text{h}$)	Peak Gradient ($^{\circ}\text{C}$)			Mean Gradient ($^{\circ}\text{C}$)	
	$\tau=2.0\text{ h}$	$\tau=2.5\text{ h}$	$\tau=3.0\text{ h}$	Min	Max
0.30	0.40	0.47	0.54	0.23	0.31
0.50	0.53	0.62	0.70	0.30	0.39
0.70	0.60	0.69	0.78	0.33	0.44
1.00	0.66	0.75	0.83	0.35	0.46
1.25	0.68	0.78	0.86	0.36	0.48
1.50	0.70	0.80	0.88	0.37	0.49
2.00	0.73	0.83	0.91	0.38	0.50

Figure 1 illustrates the tradeoff between tracking performance and gradient penalties across three operational scenarios.

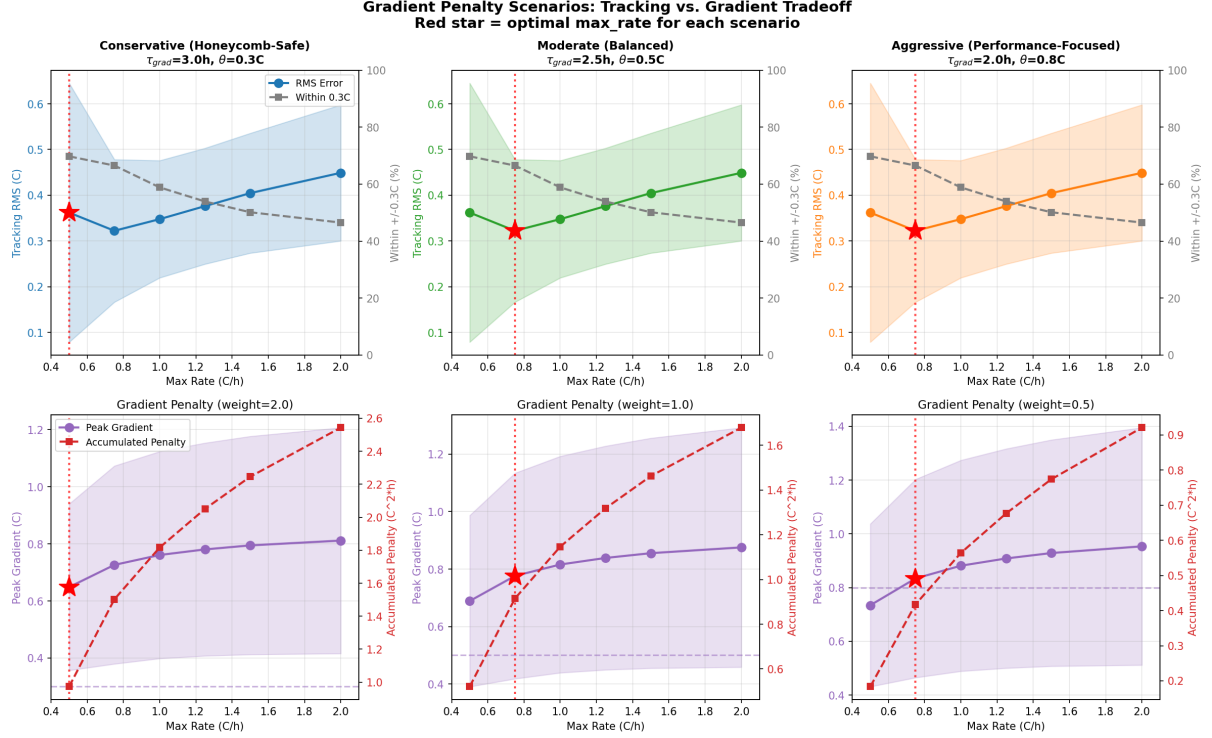


Figure 1: Tracking performance (top row) and gradient penalties (bottom row) for three operational scenarios: Conservative (honeycomb-safe), Moderate (balanced), and Aggressive (performance-focused). Red stars indicate optimal rate for each scenario. The shaded regions represent the interquartile range across 392 test nights.

5 Safety Analysis: M1M3 Gradient Limits

CRITICAL SAFETY CONSIDERATION

Based on engineering estimates, the M1M3 mirror can tolerate internal gradients in the range of 0.5 °C to 1.0 °C. Table 4 provides a safety assessment for each rate limit configuration.

Table 4: Safety assessment matrix for M1M3 internal gradients

Max Rate (°C/h)	Peak Gradient ($\tau=2.0$ h)	Peak Gradient ($\tau=3.0$ h)	Safety Assessment
0.30	0.40 °C	0.54 °C	SAFE – Below 0.5 °C threshold
0.50	0.53 °C	0.70 °C	SAFE – Within acceptable range
0.70	0.60 °C	0.78 °C	CAUTION – Mid-range, monitor closely
1.00	0.66 °C	0.83 °C	CAUTION – Approaching upper limit
1.25	0.68 °C	0.86 °C	CAUTION – Near upper limit
1.50	0.70 °C	0.88 °C	WARNING – May exceed limits
2.00	0.73 °C	0.91 °C	WARNING – Likely exceeds limits

5.1 Tracking Performance Thresholds

Table 5 shows the percentage of overnight operation time that tracking error (mirror temperature minus target) remains below critical thresholds. This directly quantifies image quality compliance.

Table 5: Tracking error compliance: percentage of time below temperature thresholds

Max Rate (°C/h)	RMS (°C)	Tracking Error Below Threshold			Gradient 95th %ile (°C)
		% <0.3 °C	% <0.5 °C	% <1.0 °C	
0.30	0.93	38.2	53.1	77.7	0.67
0.50	0.62	55.1	71.6	90.7	0.90
0.70	0.49	63.7	81.4	95.2	0.95
1.00	0.40	68.7	87.3	97.4	0.97
1.25	0.36	70.2	89.2	98.1	0.97
1.50	0.34	70.7	90.1	98.5	0.98
2.00	0.32	71.1	90.8	98.9	0.98

Note: Highlighted row indicates current operating point (0.7 °C/h). Gradient 95th percentile shown for $\tau_{\text{gradient}} = 3.0 \text{ h}$.

Key findings from tracking threshold analysis:

- At 0.7 °C/h: Tracking error stays below 0.5 °C for **81.4%** of the time, and below 1.0 °C for **95.2%** of the time
- Increasing to 1.0 °C/h improves <0.5 °C compliance from 81% to 87%, with 95th percentile gradient increasing only slightly from 0.95 to 0.97 °C
- At 0.3 °C/h: Only 53% of time below 0.5 °C and 78% below 1.0 °C — significant tracking degradation
- The 95th percentile gradient stays below 1.0 °C for all rate configurations, indicating typical operation is within safety limits

5.2 Worst-Case Analysis

For conservative honeycomb thermal behavior ($\tau_{\text{gradient}} = 3.0 \text{ h}$):

- **Current rate (0.7 °C/h):** Peak gradient = 0.78 °C
 - Within the 0.5 °C–1.0 °C acceptable range
 - However, at 78% of the upper limit
 - Recommend monitoring for signs of image quality degradation
- **If gradient tolerance is strict (<0.5 °C):**
 - Must reduce rate to 0.30 °C/h
 - Tracking performance degrades significantly: 35% vs 59% within tolerance
 - Not recommended unless gradient damage is confirmed

5.3 Maximum Observed Gradients

The simulation also tracked maximum gradient events across all nights. For worst-case nights with rapid ambient temperature changes:

Table 6: Maximum gradient events (worst-case nights)

Max Rate ($^{\circ}\text{C}/\text{h}$)	$\tau=2.0\text{ h}$	$\tau=2.5\text{ h}$	$\tau=3.0\text{ h}$	
0.30	0.59	0.72	0.86	
0.50	0.98	1.20	1.42	
0.70	1.30	1.58	1.85	<i>Note: These represent rare extreme events, not typical operation.</i>
1.00	1.63	1.96	2.26	
1.50	2.12	2.50	2.84	
2.00	2.48	2.89	3.27	

Important: Even at the current $0.7^{\circ}\text{C}/\text{h}$ rate, worst-case nights can produce gradients of 1.3°C – 1.9°C , which **exceed the 1.0°C upper limit**. This suggests that:

1. The rate limit alone may not be sufficient protection
2. Adaptive rate limiting based on ambient conditions may be necessary
3. Real-time gradient monitoring should be considered

6 Scenario Comparison

We evaluated three operational scenarios representing different risk tolerances:

Table 7: Three operational scenarios and optimal rate selection

Scenario	Description	τ_{grad}	Penalty Weight	Optimal Rate
Conservative	Honeycomb-safe, worst-case	3.0 h	2.0	$0.50^{\circ}\text{C}/\text{h}$
Moderate	Balanced tradeoff	2.5 h	1.0	$0.75^{\circ}\text{C}/\text{h}$
Aggressive	Performance-focused	2.0 h	0.5	$0.75^{\circ}\text{C}/\text{h}$ – $1.0^{\circ}\text{C}/\text{h}$

Key finding: The current operating point of $0.7^{\circ}\text{C}/\text{h}$ is validated as near-optimal for the moderate scenario, confirming it represents a reasonable engineering tradeoff.

Figure 2 shows the combined objective function (tracking RMS + weighted gradient penalty) for each scenario. The optimal rate is marked with a star.

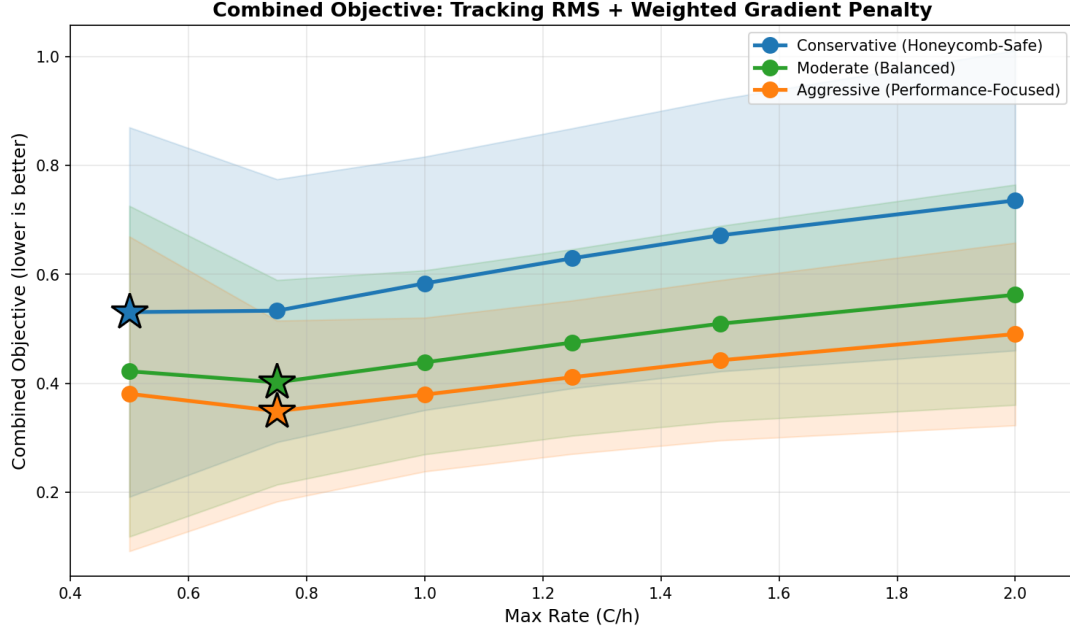


Figure 2: Combined objective (tracking RMS + weighted gradient penalty) as a function of maximum rate for three scenarios. Lower is better. Stars indicate optimal operating points. The Conservative scenario favors 0.5°C/h , while Moderate and Aggressive scenarios favor 0.75°C/h .

6.1 Pareto Frontier Analysis

Figure 3 shows the Pareto frontier—the set of non-dominated solutions in the tracking-vs-gradient tradeoff space. All tested configurations lie on or near this frontier, indicating that improving one objective necessarily degrades the other.

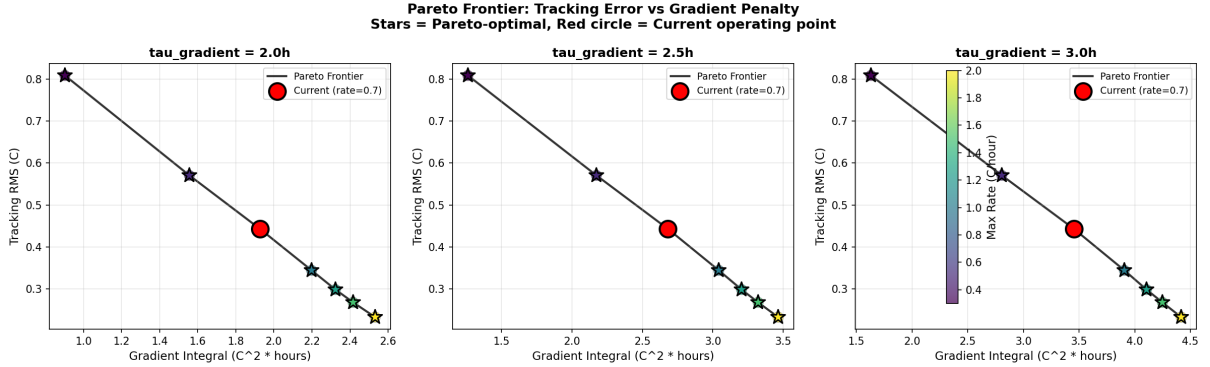


Figure 3: Pareto frontier showing tracking RMS vs. gradient integral for three values of τ_{gradient} . Stars indicate Pareto-optimal points, colored by max rate. Red circles mark the current operating point (0.7°C/h). Moving along the frontier trades tracking performance for gradient safety.

7 Recommendations

7.1 Immediate Actions

1. **Maintain current rate limit:** The 0.7°C/h rate limit produces gradients within the acceptable 0.5°C – 1.0°C range under typical conditions.
2. **Monitor image quality:** Correlate focal plane granularity measurements with thermal

rate-of-change events to validate the gradient model.

3. **Flag high-gradient nights:** Implement alerts when predicted ambient rate exceeds $0.5\text{ }^{\circ}\text{C/h}$, as these nights may produce gradients exceeding $1.0\text{ }^{\circ}\text{C}$.

7.2 If Gradient Damage is Observed

If image quality analysis reveals gradient-induced degradation:

1. Reduce rate limit to $0.5\text{ }^{\circ}\text{C/h}$ (conservative scenario)
2. Accept reduced tracking performance (48% vs 59% within tolerance)
3. Consider adaptive rate limiting (see Future Work)

7.3 If Better Tracking is Required

If operational requirements demand improved tracking performance:

1. Increasing to $1.0\text{ }^{\circ}\text{C/h}$ would improve tracking from 59% to 69%
2. Peak gradients would increase to $0.83\text{ }^{\circ}\text{C}$ (still within limits)
3. Requires validation that gradient tolerance is not exceeded

8 Future Work

8.1 Critical Parameter Validation

The following parameters require empirical measurement:

1. **Gradient relaxation time constant** (τ_{gradient}):
 - Current estimate: 2.0 h–3.0 h
 - Method: Install surface and bulk temperature sensors, measure response to step changes
 - Impact: Determines whether $0.7\text{ }^{\circ}\text{C/h}$ or $0.5\text{ }^{\circ}\text{C/h}$ is appropriate (see Figure 4)
2. **Gradient-to-image-quality mapping:**
 - Correlate PSF degradation with measured/estimated gradients
 - Establish quantitative threshold for acceptable gradient magnitude
3. **Honeycomb thermal pattern characterization:**
 - Map spatial temperature distribution during transients
 - Identify worst-case thermal response regions

Figure 4 illustrates the critical dependence of optimal rate selection on the assumed τ_{gradient} and penalty weight. This highlights the importance of empirical validation.

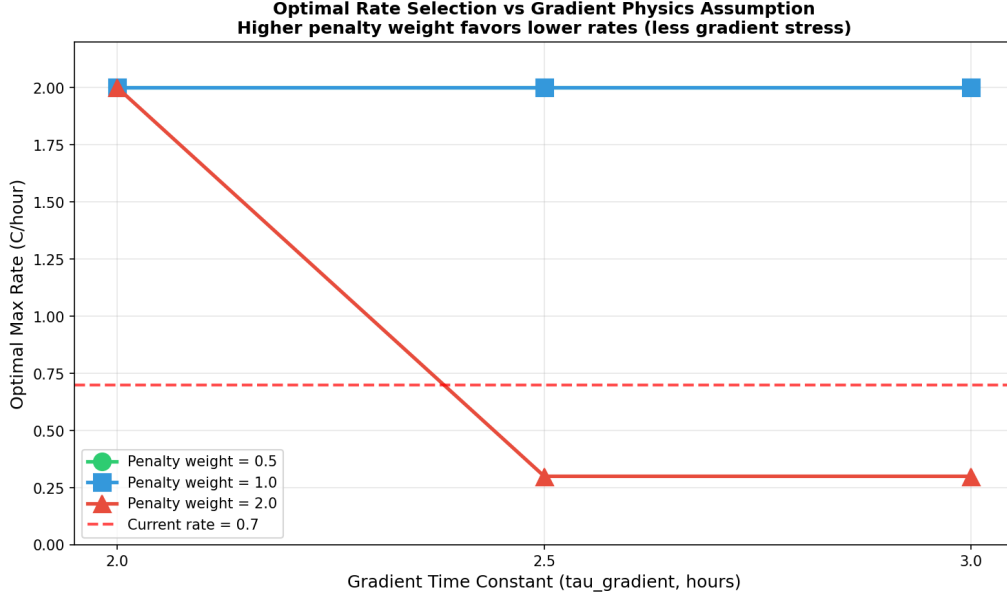


Figure 4: Optimal maximum rate as a function of gradient time constant (τ_{gradient}) for three penalty weight scenarios. At high penalty weight (red triangles), the optimal rate drops dramatically from 2.0°C/h to 0.3°C/h when $\tau_{\text{gradient}} \geq 2.5$ h. The current operating rate of 0.7°C/h (dashed line) is shown for reference.

Figure 5 shows the full optimization landscape as contour plots, revealing how the combined objective varies across the parameter space.

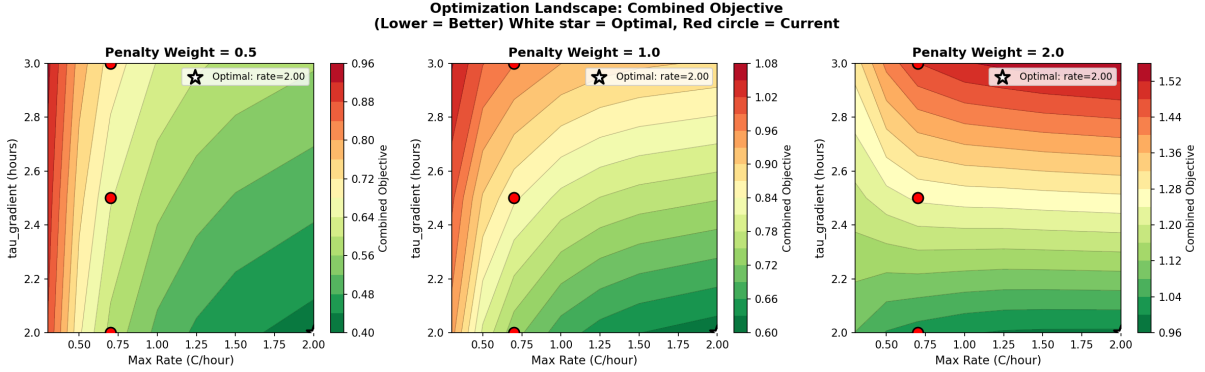


Figure 5: Optimization landscape showing combined objective (lower is better) across max rate and τ_{gradient} for three penalty weights. Green regions indicate better performance. White stars mark optimal configurations; red circles mark current operating point. Left: low penalty favors high rates. Right: high penalty creates complex landscape with optimum shifting to low rates.

8.2 Adaptive Rate Limiting

The current fixed rate limit may be suboptimal. Future development should consider:

$$r_{\text{max}}(t) = r_{\text{base}} + k \cdot \min(r_{\text{ambient}}(t), r_{\text{threshold}}) \quad (6)$$

where $r_{\text{ambient}}(t)$ is the current ambient temperature rate of change. This would:

- Allow faster tracking when ambient is changing rapidly (accepting some gradient)
- Maintain conservative rates during stable conditions

- Require real-time gradient estimation or measurement

8.3 Multi-Zone Thermal Model

The two-zone model is a simplification. A more accurate model would include:

- Multiple radial and vertical zones
- Honeycomb cell-level thermal dynamics
- Stress-optical coupling for direct image quality prediction

8.4 Integration with Weather Forecasting

The existing `rubin-twilight-forecast` temperature prediction system ($\sim 0.67^\circ\text{C}$ RMSE at 3-hour lead time) could be integrated to:

- Anticipate rapid ambient changes and pre-condition the mirror
- Optimize the daytime setpoint based on predicted sunset temperature
- Provide advance warning of high-gradient risk nights

9 Conclusions

This simulation study provides quantitative guidance for M1M3 thermal control rate limit selection:

1. The current rate limit of $0.7^\circ\text{C}/\text{h}$ is **validated** as a reasonable engineering choice, achieving 58.6% tracking within $\pm 0.3^\circ\text{C}$ while maintaining typical gradients of 0.6°C – 0.8°C .
2. **Peak gradients are within the 0.5°C – 1.0°C acceptable range** for typical nights, but worst-case nights can exceed 1.0°C even at the current rate.
3. **The critical unknown is τ_{gradient}** : If gradient relaxation is slow (3.0 h), the current rate produces gradients at 78% of the upper safety limit.
4. **Reducing to $0.5^\circ\text{C}/\text{h}$** would provide additional safety margin but degrades tracking performance from 59% to 48% within tolerance.
5. **Increasing to $1.0^\circ\text{C}/\text{h}$** would improve tracking to 69% but pushes gradients closer to the 1.0°C limit.

Recommended next step: Empirically measure τ_{gradient} through controlled thermal transient experiments to resolve the uncertainty in optimal rate selection.

A Simulation Code and Data

The simulation code and results are available in the `RubinThermal` repository:

- Two-zone model: `rubin_thermal/gradient_model.py`
- Scenario analysis: `scripts/gradient_scenarios.py`
- Pareto analysis: `scripts/pareto_analysis.py`
- Results data: `results/pareto_results.csv`
- Figures: `figures/gradient_scenarios_*.png`, `figures/pareto_*.png`